STUDENT ACTIVITY

AI Systems & Technologies — Activity

Activity: System Classification and Capability Analysis

Q: Read the four system descriptions below and classify each as Rule-based AI, Machine Learning, Deep Learning, or General Artificial Intelligence.

Next, describe the differences in data dependency and human intervention between these systems in four to five sentences, ensuring you mention at least one limitation of AI.

The descriptions:

- A system that analyzes patterns in data to filter spam without being explicitly programmed for every single task.
- A system that uses artificial neural networks with multiple hidden layers to recognize faces in complex unstructured data.
- A system that does not involve learning from data but relies entirely on significant human-defined rules and logic.
- A system designed to simulate human intelligence and operate continuously with 24/7 availability without fatigue.

Classification of the Systems

System Description	Classification
A system that analyzes patterns in data to filter spam without being explicitly programmed for every single task.	**Machine Learning (ML)**
A system that uses artificial neural networks with multiple hidden layers to recognize faces in complex unstructured data.	**Deep Learning (DL)**
A system that does not involve learning from data but relies entirely on significant human-defined rules and logic.	**Rule-based AI**
A system designed to simulate human intelligence and operate continuously with 24/7 availability without fatigue.	**General Artificial Intelligence (General AI / AGI)**

Differences in Data Dependency and Human Intervention

Rule-based AI depends on human-created rules and does not learn from data, requiring significant human intervention to update its knowledge. Machine Learning learns patterns from data, reducing the need for manual programming but still requiring humans to prepare data and tune models. Deep Learning relies on large volumes of data and multiple neural network layers, requiring less manual feature engineering but greater computational resources. General Artificial Intelligence aims to perform a wide variety of human-like tasks independently with minimal human intervention, although true AGI has not yet been achieved. One important limitation of AI is that its performance depends heavily on the quality of data, and it may produce biased or inaccurate results if trained on poor or biased datasets.